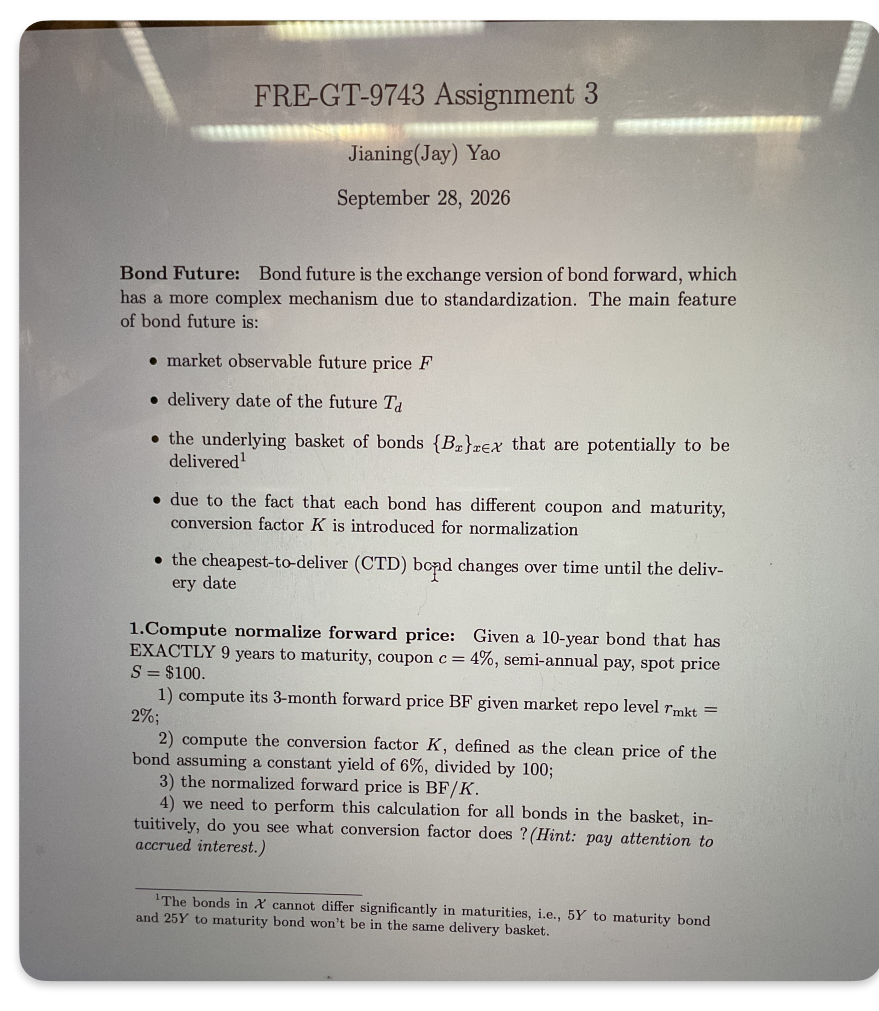


Homework 3

Question 1:



Solution:

10 year bond
9 years to maturity
 $c = 4\%$
paid semi-annually $\Rightarrow$ twice a year
 $s = \$100$

① $r_{\text{mkt}} = 2\%$

borrow money today $\rightarrow$ buy the bond $\rightarrow$ hold 3 months $\rightarrow$ pay back loan
How much do I pay to buy bond today:

Price = Quoted Price + Accrued Interest

$AI = 0$ (since exactly 9 yrs left)

$\therefore$ Price Today = \$100

How much owed on loan after 3 months:

borrow \$100 to buy bond

$r = 2\%$ per year

borrowing for 3 months

$\frac{3}{12} = 0.25$ years

Interest = Amt borrowed $\times$ Rate $\times$ Time (yrs)

$= 100 \times 0.02 \times 0.25$

$= 0.50$

total owed after 3 months = $100 + 0.50$

$= \$100.50$

Did bond pay anything in 3 months?

coupons come every 6 months

we hold for 3 months

coupons received during 3m = 0

$\therefore$ dirty forward price = $\boxed{\$100.50}$

Removing accrued interest at 3 month point

clean price = dirty price - AI

AI in 3m = $\frac{\text{full coupon} \times \text{time held}}{6m}$

$= \$2 \times \frac{3}{6}$

$= \$1$

$\therefore$ Clean forward price = $\$100.50 - \1

$= \boxed{\$99.50}$ Answer

paying \$0.50 of interest
earning \$1 of bond interest
 $\therefore$ \$0.50 ahead
Holding bond is profitable
 $\therefore$ forward price is \$0.50 lower than today's price.

2. If market demanded 6% return on hand, what would price be?

bond's price is basically all the money it pays you discounted

Finding rate per period:

Rate per 6 months = $6\% \div 2$

$= 3\%$

$= 0.03$

counting periods:

no. of 6 month periods = 9×2

$= 18$

Rule for shrinking a future payment:

Value today = $\frac{\text{Payment}}{(1.03 \times 1.03 \times \dots \times 1.03)}$

one for every period waiting

$= \frac{100}{(1.03)^{18}}$

$= \$58.739$

value today of all 18 coupons of \$2 = $2 \times 0.41261 \div 0.03$

$= 2 \times 13.7535$

$= 27.507$

Price at 6% yield = $27.507 + 58.739$

$= \$86.2465$

$\boxed{0.862465}$ Answer

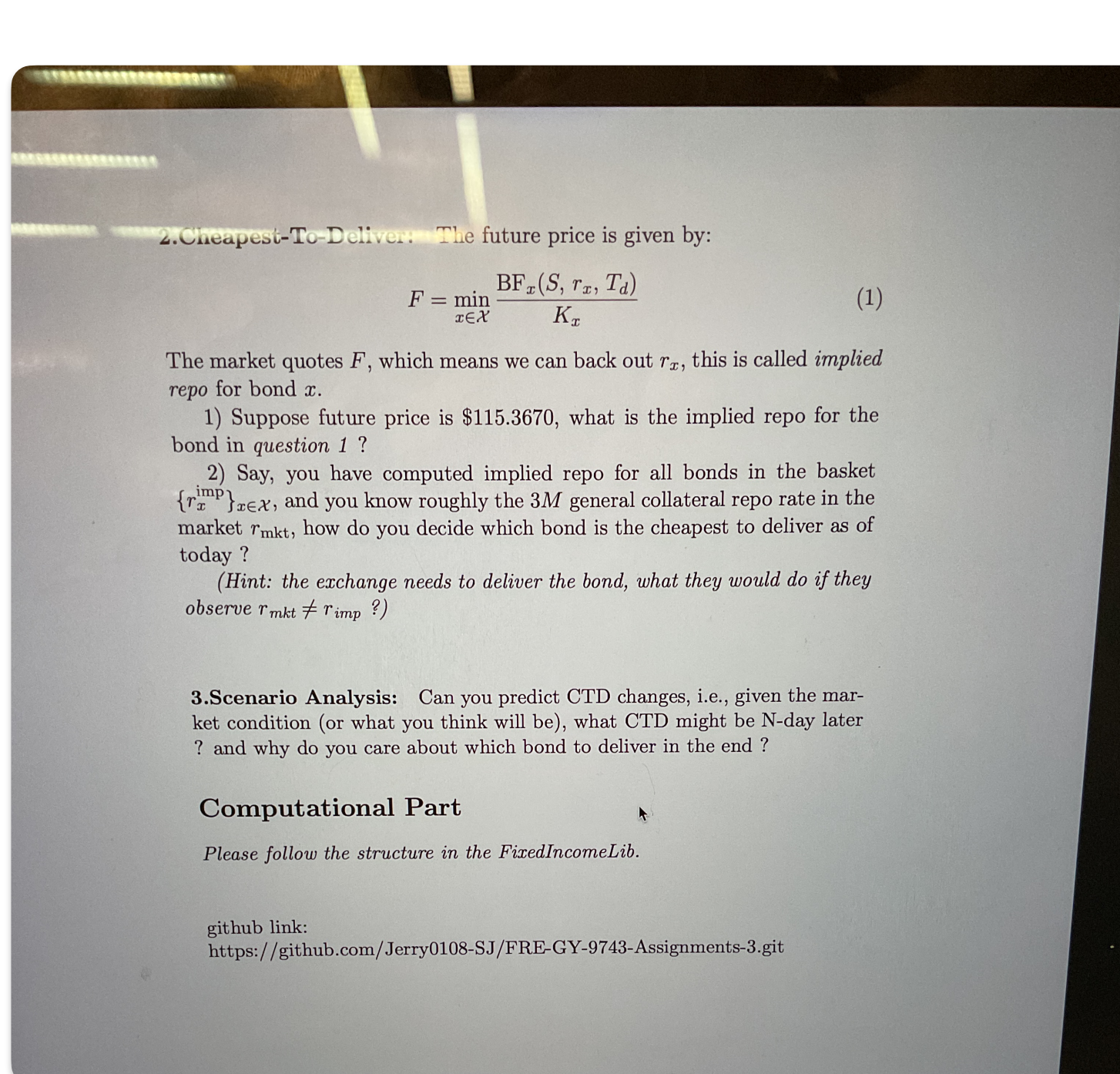
3. Normalized forward price

$\frac{BF}{K} = \frac{99.5}{0.862765}$

$= \boxed{\$115.367}$ Answer

4. The conversion factor normalizes each bond in the basket to a standard 6% coupon bond, so bonds with different coupons and maturities can be compared on the same scale. Low coupon bonds have $K < 1$ and high coupon bonds have $K > 1$. K is based on the clean price, so it adjusts only for coupon and maturity differences, while accrued interest is paid separately at delivery. That is why we stripped the accrued interest out of the forward before dividing by K .

Question 2



$$F = \min_{x \in X} \frac{BF_x(S, r_x, T_d)}{K_x}$$

F = future price on market screen
 x = one bond from the basket
 BF_x = that bond's forward price
 K_x = conversion factor

1. $BF = F \times K$

$= 115.3670 \times 0.862465$

$= 99.50$

dirty forward = $99.50 + 1$

$= 100.50$

$100 \times (1 + r \times 0.25) = 100.50$ (from prev ques)

$1 + r \times 0.25 = 1.0050$

$r \times 0.25 = 0.0050$

$r = \frac{0.0050}{0.25}$

$= 0.02$

$= \boxed{2\%}$ Answer

2. The implied repo is the return earned by buying a bond today, financing it at repo, selling the future, and delivering the bond (and 3 coupons). The short chooses which bond to deliver and will pick the one that earns the most, so the CTD is the bond with the highest implied repo (equivalently the lowest BF/K).

If $r_{\text{imp}} > r_{\text{mkt}}$ for some bond, there's an arbitrage: borrow at r_{mkt} , buy the bond, short the future, and deliver, selling in a riskless profit. This selling pressure pushes F down until $r_{\text{imp}} \leq r_{\text{mkt}}$.

If $r_{\text{imp}} < r_{\text{mkt}}$: If $r_{\text{imp}} < r_{\text{mkt}}$, cash and carry loses money.

So in equilibrium the CTD's implied repo is close to the market 4% repo rate, and all other bonds have lower implied repos. For the bond in Q1, $r_{\text{imp}} = 2\%$, $r_{\text{mkt}} = 2\%$. So it is the CTD.

Question 3

3. Scenario Analysis: Can you predict CTD changes, i.e., given the market condition (or what you think will be), what CTD might be N-day later?

? and why do you care about which bond to deliver in the end?

Predicting

Yes, the CTD can be predicted conditionally on a market scenario. Since the CTD is the bond with the lowest BF/K and K is fixed (computed at a 6% yield) while bond prices still move with the market, the CTD changes when relative bond prices change. The main driver is the yield level relative to 6%: when yields are above 6%, high duration bonds (longer maturity, lower coupon) fall more in price and become CTD. When yields are below 6%, low duration bonds (shorter maturity, higher coupon) become CTD. Near 6% bonds are close and the CTD can switch easily. Other drivers include the curve shape (steepening favours long bonds as CTD, flattening favours short bonds), bond specific repo (specialness), carry and the passage of time to delivery. To predict the CTD N days later, I would define scenarios (parallel shifts, steepening/flattening, 1 bp changes), reprice every bond in the basket under each scenario, recompute each BF with the remaining time to delivery, divide by the fixed K , and identify the bond with the lowest BF/K (highest implied repo) in each scenario.

$\rightarrow$ the future's price tracks the CTD, so the CTD determines how the future moves.
 $\rightarrow$ Hedger's ratio depends on the CTD's DV01 divided by its conversion factor, so a CTD switch changes the hedge ratio.
 $\rightarrow$ The short's right to choose the delivered bond is a switch option whose value depends on the probability of CTD changes, and it makes the future trade below the plain vanilla BF/K .

$\rightarrow$ Basis traders trade the spread between bonds and futures based on CTD expectations.

$\rightarrow$ The short must actually source and deliver the bond so availability and repo specialness of the likely CTD matter.